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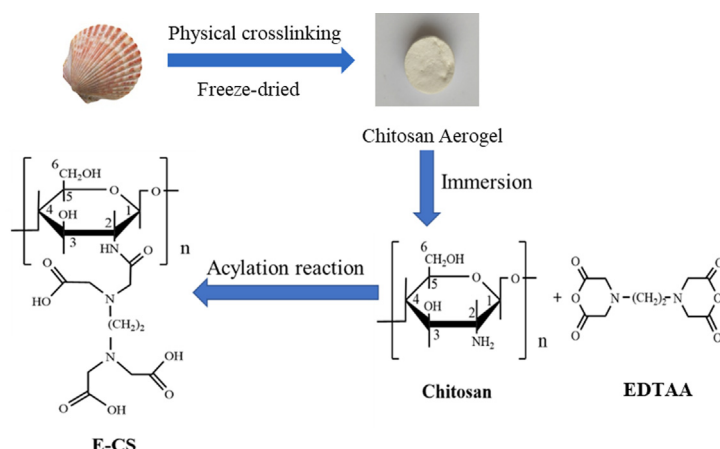
A 'top modification' strategy for enhancing the ability of a chitosan aerogel to efficiently capture heavy metal ions

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ABSTRACT

Chitosan is a promising substitute for heavy metal ion adsorbents. However, traditional pure chitosan adsorbents have certain disadvantages that limit their application. In this paper, a 'top modification' strategy was used to enhance the capturing ability of chitosan adsorbents. A chitosan aerogel was prepared via physical crosslinking and then enhanced by immersion in ethylenediamine tetraacetic anhydride solution. Finally, an enhanced chitosan aerogel was obtained, and analyses were used to describe its structure, adsorption properties and mechanism. Results showed that both the porous structure and the combined complexations dramatically improved the capturing ability of the chitosan aerogel for heavy metal ions. The theoretical adsorption capacities of the enhanced aerogel for Cu^{2+} , Pb^{2+} and Cd^{2+} reached 108.14, 143.73 and 84.62 mg/g, respectively. Due to their environmental friendliness, good adsorption performance, easy separation and reusability, enhanced aerogels have become viable solutions to removing heavy metal pollutants from aquatic systems.

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1. Introduction

Heavy metals, such as copper, lead and cadmium, are emitted by human activities, such as burning fossil fuels, mining and metallurgy, and enter aquatic systems through various pathways.

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Currently, heavy metal ions have become a primary factor that causes water pollution. These heavy metal ions are not degradable and tend to accumulate in biological tissues via the food chain in water ecosystem systems, causing serious diseases and harming human health [1]. Methods to reduce heavy metal ions in water include adsorption [2], precipitation [3], nanofiltration [4] and oxidation [5]. Adsorption is considered a promising water treatment technology due to its simple operation, low energy consumption, low cost, and renewable adsorbents [6]. Mineral adsorbents [7] and nanoparticle adsorbents [8] have been studied extensively; however, most exhibit low reusability and cause secondary harm to the environment. To achieve efficient treatment of wastewater and reduce secondary pollution, researchers have been investigating environmentally friendly chitosan (CS) adsorbents [9–16].

Chitosan is the only alkaline polysaccharide in nature and contains many hydroxyl groups and amino groups that have high activities for heavy metal ions [17]. However, traditional pure chitosan adsorbents exhibit low porosity and are easily lost and unstable in acidic media, limiting their application [18]. The rich porous structure of aerogels can be used to compensate for these defects. Aerogels with high porosity capture heavy metal ions more easily. Concurrently, compared to powdery chitosan adsorbents, blocky aerogels are more promising because they can be easily separated and recovered from aquatic systems. However, chemical crosslinking, which is traditionally used to prepare CS aerogels [9,19,20], typically reduces the number of active sites for heavy metal ions, which is a primary reason for the low capturing ability of pure chitosan adsorbents. To improve adsorption efficiency, researchers constructed chitosan hydrogels via physical crosslinking [21]. This method effectively mitigates the consumption of adsorption sites; however, weak physical interactions lead to lower structural stability, which limits its reusability. In this study, to enhance the capturing ability of pure chitosan adsorbent, ethylenediamine tetraacetic anhydride (EDTAA) is used to increase the number of active sites via a simple immersion method after blocky chitosan aerogel was prepared via physical crosslinking. The investigation was performed on the structure, adsorption properties and mechanism of the enhanced chitosan aerogel. This study provides a relatively easy and efficient ‘top modification’ strategy to create an enhanced pure chitosan adsorbent for heavy metal ions.

2. Materials and methods

2.1. Materials

Chitosan (CS) ($M_w = 3.0 \times 10^5$ with 90% deacetylation degree, biological reagent) was purchased from Sinopharm Chemical Reagent Co., Ltd. Ethanol was received from Xilong Chemical Co., Ltd., China. Ethylenediamine tetraacetic anhydride (EDTAA) and dimethyl sulfoxide (DMSO) were purchased from McLean Reagent Company. Urea, lithium hydroxide and potassium hydroxide were supplied by Guanghua SciTech Co., Ltd., China. $\text{Cu}(\text{NO}_3)_2$, $\text{Cd}(\text{NO}_3)_2$ and $\text{Pb}(\text{NO}_3)_2$ were supplied by Xilong Chemical Co., Ltd., China. All chemicals were analytic grade and used as received without further purification unless otherwise stated. Deionized water was used throughout the experimental process.

2.2. Preparation of chitosan aerogel

Chitosan (0.2 g) was dispersed into 10 ml alkaline solution containing $\text{LiOH/KOH/urea/H}_2\text{O}$ in a mass ratio of 4.5:7.0:8.0:80.5. After multiple freezing/thawing cycles, a transparent chitosan solution was obtained, and the solution was molded in a culture dish. A stable hydrogel was obtained after standing at 30 °C for

12 h. Then, to remove residues, the hydrogel was soaked in deionized water, which was replaced every 5 h until pH = 7. The hydrogel was frozen at –18 °C for 24 h before being freeze-dried for 48 h. Finally, a CS aerogel was obtained.

2.3. Enhancement of chitosan aerogel

Following the immersion method, the prepared CS aerogel was immersed in 8 ml DMSO solution with 0.6 g EDTAA. The reaction was performed at 70 °C for 12 h. Then, the product was thoroughly rinsed with ethanol and deionized water to remove unreacted reagent. After lyophilization, the enhanced sample was obtained and referred to as E-CS aerogel. The percentage graft yield (GY, %) was calculated according to the following formula:

$$\text{GY} = (M_2 - M_1)/M_1 \times 100\% \quad (1)$$

where M_1 and M_2 are the weights of the CS aerogel and E-CS aerogel, respectively.

2.4. Characterizations

The micromorphology of the samples was observed via scanning electron microscopy (SEM, ZEISS EVO18, Germany). Changes in functional groups were determined by a Fourier transform infrared spectrometer (FTIR, Bruker, model Tensor 27, Germany). Changes in binding energy were observed by X-ray photoelectron spectroscopy (XPS, Thermo Scientific Escalab 250Xi, USA). Thermal stability analysis was performed on a simultaneous thermal analyzer (TGA, JingYiGaoKe, ZCT-1, China) under a dynamic nitrogen atmosphere with a heating rate of 10 °C/min.

The apparent density was expressed in the following formula:

$$\rho_{\text{CSA}} = \frac{m}{V} \quad (\text{g}/\text{cm}^3) \quad (2)$$

where m and V are the weight and volume of the aerogel, respectively.

The porosity (P) was expressed in the following formula:

$$P = \left(1 - \frac{\rho_{\text{CSA}}}{\rho_s}\right) \times 100\% \quad (3)$$

where ρ_{CSA} and ρ_s are the densities of the aerogel and solid scaffold (approximated to a chitosan skeleton of 1.46 g/cm^3), respectively.

2.5. Adsorption experiment for heavy metal ions

$\text{Cu}(\text{NO}_3)_2$, $\text{Cd}(\text{NO}_3)_2$ and $\text{Pb}(\text{NO}_3)_2$ were used to prepare the Cu (II), Cd(II), and Pb(II) stock solutions. The pH values of these solutions were adjusted by 0.1 M HCl or NaOH solutions. The adsorption experiments were performed in a thermostatic oscillator (Changzhou PuTian, SHA-CA, China) with a desired temperature of 298 K and a fixed rotation speed of 150 rpm. The concentrations of heavy metal ions in the solutions were monitored on an atomic absorption spectrophotometer (Beijing PuXi, TAS-990, China).

The adsorption capacity (Q , mg/g) and removal efficiency (R , %) were calculated according to formulae (4) and (5) [22]:

$$Q = V(C_0 - C_t)/m_0 \quad (4)$$

$$R = (C_0 - C_e)/C_0 \times 100\% \quad (5)$$

where C_0 (mg/L), C_t (mg/L) and C_e (mg/L) are heavy metal ion concentrations at the initial time, given time and adsorption equilibrium time, respectively; V (L) is the volume of heavy metal ion solution; and m_0 (g) is the weight of the adsorbent.

The effect of the pH of the aqueous solution on adsorption was investigated by oscillating at 150 rpm for 12 h with pH values from

2 to 7. In this process, 0.1 g E-CS aerogel was placed in 100 ml 50 mg/L heavy metal ion solution.

To investigate adsorption kinetics, 0.2 g E-CS aerogel was placed in 200 ml 50 mg/L heavy metal ion solution at pH = 5 for a given duration. The adsorption kinetics of the E-CS aerogel were evaluated using the pseudo first-order kinetic equation (6) and pseudo second-order kinetic equation (7) [23,24]:

$$Q_t = Q_e(1 - \exp(-k_1 t)) \quad (6)$$

$$Q_t = (k_2 Q_e^2 t) / (1 + k_2 Q_e t) \quad (7)$$

where Q_e (mg/g) is the fitting value of equilibrium adsorption capacity, t (h) is the adsorption time, Q_t (mg/g) is the adsorption capacity at given time t , and k_1 and k_2 are adsorption rate constants of the pseudo first-order kinetic equation and pseudo second-order kinetic equation, respectively.

Isothermal adsorption was tested by placing 0.1 g E-CS aerogel in a heavy metal ion solution with pH = 5 for 12 h. The data were fitted by both the Langmuir model (8) [25] and the Freundlich model (9) [26]. The saturated adsorption capacity (Q_m) was obtained from the Langmuir model:

$$Q_e = K_L Q_m C_e / (1 + K_L C_e) \quad (8)$$

$$Q_e = K_F C_e^{1/n} \quad (9)$$

where Q_e (mg/g) is the equilibrium adsorption capacity, C_e (mg/L) is the equilibrium concentration, K_L is the Langmuir adsorption constant, and K_F and $1/n$ are Freundlich adsorption constants.

2.6. Regeneration of E-CS aerogel

To evaluate the regeneration ability of the E-CS aerogel, the metal ion-loaded samples were desorbed and regenerated with the pickling method [27].

E-CS aerogel (0.1 g) was added to a conical flask containing 100 ml heavy metal ion solution (50 mg/L), and the solution was shaken at 150 rpm for 12 h. Then, the E-CS aerogel was removed to measure the removal efficiency. Then, the metal ion-loaded E-CS aerogel was placed in a conical flask containing 100 ml 0.5 M HCl solution and shaken at 150 rpm for 12 h to regenerate. The refreshed E-CS aerogel was obtained after freeze-drying again and was used to measure the removal efficiency in the next cycle.

3. Results and discussion

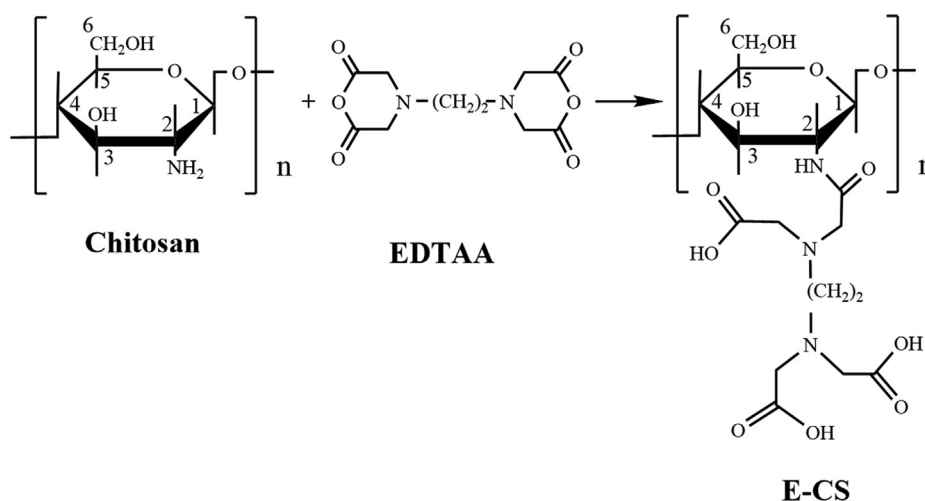
3.1. Enhancement mechanism of ethylenediamine tetraacetic anhydride for chitosan aerogel

As a promising candidate for heavy metal ion adsorbents, chitosan molecules contain many hydroxyl groups and amino groups with high activity for heavy metal ions. There are two ways to build CS aerogel's structure: chemical crosslinking and physical crosslinking. Typically, to form a stable 3D network structure, chemical reactions occur between crosslinking agents and active groups of chitosan during chemical crosslinking. These reactions consume many active groups of the chitosan, which play a major role in capturing heavy metal ions and lead to degradation of the adsorption properties. In this study, physical crosslinking was used to retain as many active groups as possible. The resulting physical-crosslinking hydrogel was created with hydrogen-bonding interactions between the chitosan molecules. While heating up in the solvent, the chitosan molecules reform, such as chains curling and winding, which can result in intensive hydrogen-bonding interactions. Under the synthetic influence of hydrophobic interactions and hydrogen-bonding interactions, a physically crosslinked hydrogel was formed without the loss of active groups. After lyophilization, a CS aerogel with as many active groups as possible was created.

To improve the capturing ability to heavy metal ions using a relatively easy and efficient 'top modification' strategy, the CS aerogel was enhanced via an acylation reaction between the amino groups of chitosan and carboxylic groups of EDTAA in this study. In theory, one amino group of the chitosan molecule could exchange for three carboxylic groups of EDTAA, as shown in Scheme 1; these groups become new active sites for heavy metal ions. Considering that one EDTAA molecule could react with more than one amino group, it was reasonable to infer that there was a bridge effect to make the aerogel's structure more stable.

3.2. Structure of E-CS aerogel

The FTIR spectra of the CS aerogel and E-CS aerogel indicated a chemical reaction between chitosan and EDTAA, as shown in Fig. 1. The wide peak from 3000 to 3700 cm^{-1} can be attributed to $-\text{OH}$ intramolecular association and $-\text{NH}_2$ stretching vibration. In the FTIR spectra of the CS aerogel, the peaks at 1590 cm^{-1} and 1652 cm^{-1} corresponded to amides I and amides II in the chitosan molecule, respectively [28]. In the FTIR spectra of the E-CS aerogel,



Scheme 1. Acylation reaction between chitosan unit and EDTAA.

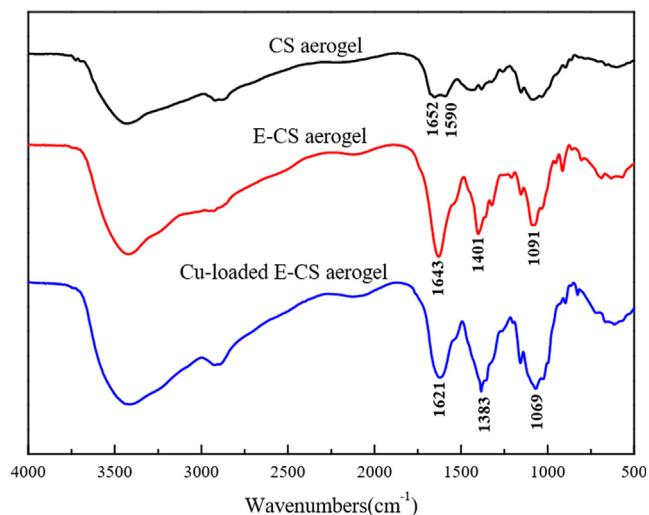


Fig. 1. FTIR spectra of CS aerogel, E-CS aerogel and Cu-loaded E-CS aerogel.

a strong characteristic peak occurred at 1401 cm^{-1} , which can be attributed to the stretching vibration of carboxyl groups, indicating that carboxyl groups were present in the structure. Concurrently, there was a sharp band at 1091 cm^{-1} , which can be attributed to the C—O stretching vibration of the carboxyl group covering the peak of the sugar ring [29]. The characteristic peak of N—C=O of the amide group was also present at 1643 cm^{-1} in the FTIR spectra of the E-CS aerogel. However, the characteristic peaks, which corresponded to esterification of the primary hydroxyl at C6 of the chitosan molecule, were not observed. Thus, the primary hydroxyl at C6 of the chitosan molecule was not involved in the reaction with EDTAA. Therefore, the acylation reaction between EDTAA

and the amino groups of the chitosan molecule occurred successfully. After loading Cu^{2+} , the C—O stretching vibration peak of the carboxyl group became stronger and wider at 1069 cm^{-1} . Thus, the carboxyl groups from EDTAA became new active sites for heavy metal ions. The calculated graft yield was 49.31%, and the molar ratio of chitosan units to EDTAA molecules was approximately 3.22:1.

The microstructures of the samples were observed via SEM, as shown in Fig. 2. Both the CS aerogel and E-CS aerogel had typical lamellar porous network structures, and EDTAA had little effect on the aerogel's microstructure. As shown in Fig. 2C, there was a secondary network structure in the lamellar structure, which was conducive to diffusion of the solution into the aerogel matrix and could promote the adsorption of heavy metal ions. Before and after enhancement, the density and porosity of the aerogel changed from 0.0257 to 0.0383 g/cm^3 and from 98.24 to 97.38%, respectively.

The thermogravimetric (TG) and differential thermogravimetric (DTG) curves of the CS aerogel and E-CS aerogel are shown in Fig. 2D. The DTG peak of CS aerogel at $293.5\text{ }^\circ\text{C}$ can be attributed to depolymerization of chitosan chains. After enhancement with EDTAA, a small peak appeared at $242.6\text{ }^\circ\text{C}$, which was caused by the thermal decomposition of EDTAA residues. The primary peak at $306.8\text{ }^\circ\text{C}$ was generated by the pyrolysis of the E-CS aerogel. Thus, the E-CS aerogel had better thermal stability, which led to better structural stability.

3.3. Adsorption properties of E-CS aerogel

Requiring a low dosage is an ideal characteristic of an adsorbent. Fig. 3A shows the influence of adsorbent dosage on the removal efficiency of Cu^{2+} , Pb^{2+} and Cd^{2+} with an initial metal ion concentration of 20 mg/L and $\text{pH} = 5$. With increasing

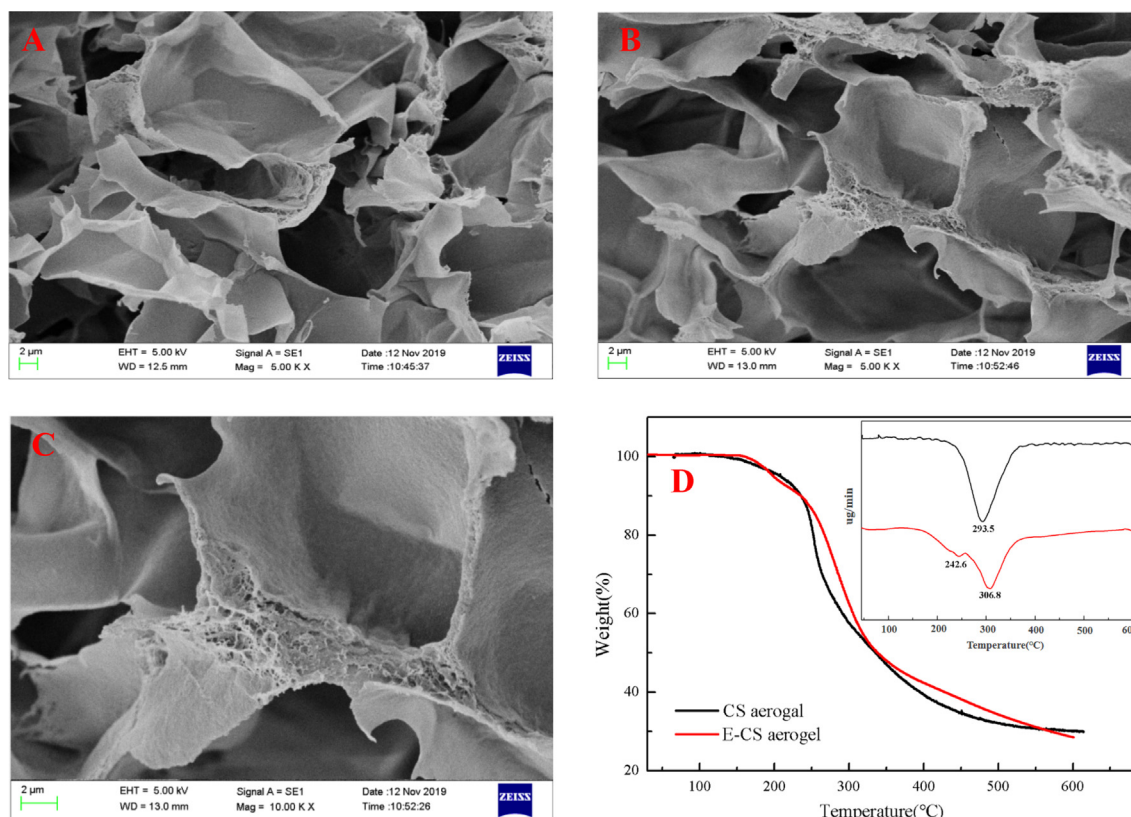


Fig. 2. SEM images of the CS aerogel (A) and E-CS aerogel (B, C) and TG curves of the CS aerogel and E-CS aerogel (inset shows the DTG curves) (D).

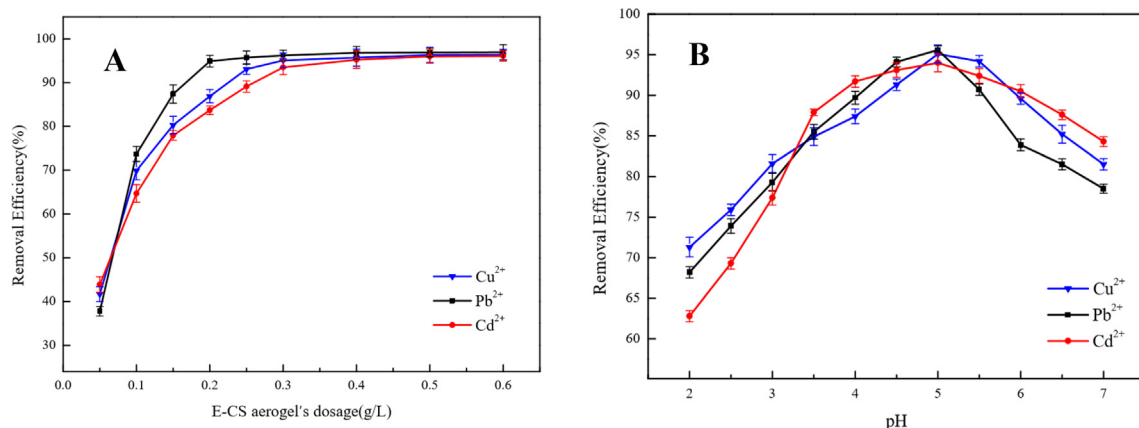


Fig. 3. Influence of E-CS aerogel dosage (A) and pH of solution (B) on removal efficiency for Cu²⁺, Pb²⁺ and Cd²⁺. Three parallel tests were performed for each sample.

adsorbent dosage, the removal efficiency for all metal ions first increased and then remained constant. The optimum dosages were 0.3, 0.2 and 0.4 g/L for Cu²⁺, Pb²⁺ and Cd²⁺, respectively. Thus, the application of E-CS aerogel for removing heavy metal ions is economical.

Due to its influence on the adsorption site and the physico-chemical state of heavy metal ions, the pH of the solution is an important factor for the adsorption properties of an adsorbent [30]. Fig. 3B shows that the removal efficiency of the E-CS aerogel for Cu²⁺, Pb²⁺ and Cd²⁺ increased with increasing solution pH. The increase in removal efficiency was the result of the reduced number of H⁺ ions with pH values from 2.5 to 5, which was attributed to the electrostatic repulsion exerted by H⁺ on metal ions. Thus, more metal ions could be adsorbed on the active sites of the adsorbent with increasing pH, and there was adsorption competition between H⁺ and metal ions. However, a turning point was found at pH 5. The removal efficiency gradually decreased due to the hydrolysis of metal ions when the pH was above 5. As shown in Fig. 4, the color of the E-CS aerogel changed from white to blue observably after adsorbing Cu²⁺. Due to its low density, the sample always floated on the surface of the solution and remained intact during oscillating adsorption. This characteristic made the E-CS aerogel easy to collect and remove from the water.

Good regeneration performance is important for an adsorbent [31]. The E-CS aerogel was successfully regenerated via the pickling method in the desorption experiment. The adsorption/desorption cycle experiment indicated that the E-CS aerogel retained

good regeneration performance at least three times. As shown in Fig. 4D, after three adsorption–desorption cycles, the removal efficiencies of the E-CS aerogel for Cu²⁺, Pb²⁺ and Cd²⁺ marginally decreased to 91.7%, 91.2% and 90.3%, respectively, which were still higher than those of the CS aerogel (84.5%, 83.3% and 80.7%). Therefore, the E-CS aerogel exhibited good regeneration performance, as well as higher adsorption capacity for heavy metal ions.

3.4. Adsorption mechanism of E-CS aerogel

To describe the adsorption process of the E-CS aerogel, the adsorption mechanism was then evaluated with the pseudo first-order kinetic model and pseudo second-order kinetic model to describe adsorption kinetics, and the Langmuir isotherm model and Freundlich isotherm model to describe adsorption thermodynamics.

The adsorption kinetics of the E-CS aerogel for three heavy metal ions were investigated at pH 5 and an ion concentration of 50 mg/L, as shown in Fig. 5A, B and C. There was a rapid adsorption stage in the first 70 min, and then the adsorption tended to reach equilibrium after 4 h. The fitting results of the kinetic models are shown in Table 1. According to the fitting results, the correlation coefficients R^2 of Cu²⁺, Pb²⁺ and Cd²⁺ were 0.964, 0.944 and 0.970 with the pseudo first-order kinetic model, respectively. The R^2 values of Cu²⁺, Pb²⁺ and Cd²⁺ were 0.985, 0.972 and 0.979 with the pseudo second-order kinetic model, respectively. The adsorption process of the E-CS aerogel for heavy metal ions was thus well

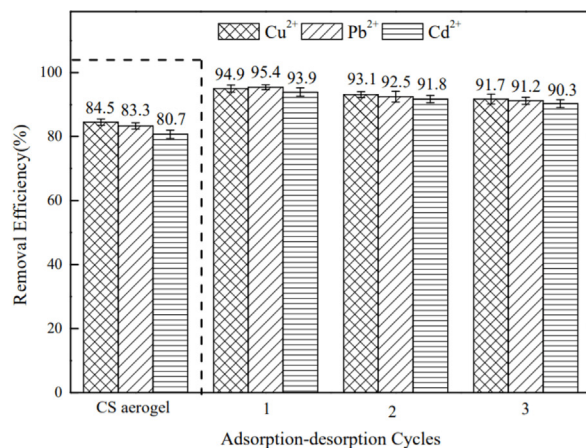
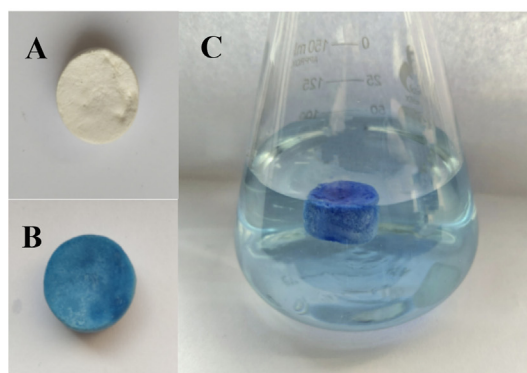


Fig. 4. E-CS aerogel before (A) and after (B) adsorbing Cu²⁺; the state of the E-CS aerogel during adsorption (C); and the removal efficiency of the E-CS aerogel in three adsorption–desorption cycles, where the dotted box indicates the removal efficiency of the CS aerogel before being modified by EDTAA (D). Three parallel tests were performed for each sample.

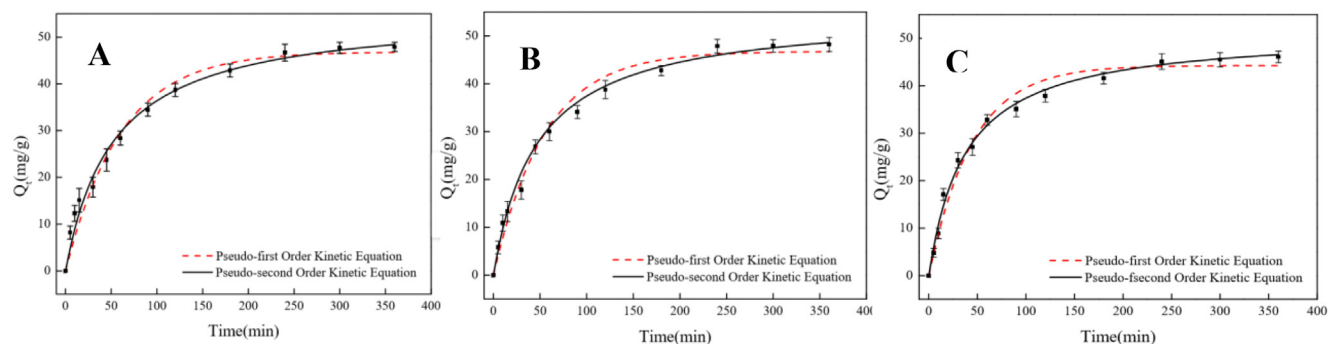


Fig. 5. Time-dependent adsorption capacity and kinetic fittings of the E-CS aerogel for Cu²⁺ (A), Pb²⁺ (B) and Cd²⁺ (C). Three parallel tests were performed for each sample.

Table 1

Fitting constants with different kinetic models for the adsorption process of E-CS aerogel.

Metal ions	Q_{exp} (mg/g)	Pseudo first order kinetic model			Pseudo second order kinetic model		
		k_1 (L/min)	Q_e (mg/g)	R^2	k_2 (g/(mg/min))	Q_e (mg/g)	R^2
Cu ²⁺	48.34	0.0243	46.82	0.964	0.001439	55.50	0.985
Pb ²⁺	48.67	0.0392	47.33	0.944	0.000764	56.55	0.972
Cd ²⁺	47.09	0.0227	44.28	0.970	0.000519	50.91	0.979

* Q_{exp} indicates the experimental equilibrium adsorption capacity.

described by a pseudo second-order kinetic equation, and therefore, it was believed that there was a chemical process of electron sharing and electron transfer between the E-CS aerogel and metal ions. According to the pseudo second-order kinetic model, the equilibrium adsorption capacities of the E-CS aerogel for Cu²⁺, Pb²⁺ and Cd²⁺ reached 55.50, 56.55 and 50.91 mg/g, respectively.

As shown in Fig. 6, the maximum adsorption efficiency was shown when the concentration of heavy metal ions reached a certain value due to the fixed number of active sites in the aerogel [32]. Both Langmuir and Freundlich isotherm models were used to perform nonlinear fitting with the experimental data, as shown in Fig. 6. Additionally, the parameters are shown in Table 2, where it appeared that the adsorbent-adsorbate system was better explained by the Langmuir model in accordance with R^2 (0.972 for Cu²⁺, 0.942 for Pb²⁺ and 0.954 for Cd²⁺). The mechanism obeyed single molecule layer adsorption or chemical adsorption, and the

adsorption was uniform [33]. According to the Freundlich model, the K_F parameters of each metal ion were different, and all values of n were greater than 1, indicating that the adsorption process might occur via electrostatic interactions, ion exchange or synergistic mechanisms [34]. Based on the Langmuir isotherm model, the calculated maximum adsorption capacities of the E-CS aerogel for Cu²⁺, Pb²⁺ and Cd²⁺ were 108.14, 143.73 and 84.62 mg/g, respectively. Thus, the E-CS aerogel exhibited excellent adsorption efficiency and was a promising adsorbent for capturing heavy metal ions.

As shown in Table 3, the adsorption performance of the E-CS aerogel was far higher than that of most chitosan-based adsorbents reported in recent years, particularly pure chitosan adsorbents. This result likely occurred due to two reasons. First, the porous structure of the aerogel had a porosity up to 97.38%, which could achieve excellent physical adsorption. Second, the increased number of active groups, as shown in FTIR spectra, allowed for efficient chemical adsorption. The adsorption capacity of the E-CS aerogel was lower than that of a wastepaper/chitosan aerogel, which had the highest adsorption capacity of 156.3 mg/g for Cu²⁺. As described in [16], a wastepaper/chitosan aerogel was prepared via pure physical crosslinking, which completely mitigated the consumption of active groups. Thus, it is important to retain as many active groups as possible for an efficient adsorbent.

The FTIR spectra of aerogels before and after adsorbing Cu²⁺ are shown in Fig. 1. For the Cu-loaded E-CS aerogel, the peaks at 1643 cm⁻¹, 1401 cm⁻¹ and 1091 cm⁻¹ moved to 1621 cm⁻¹, 1383 cm⁻¹ and 1069 cm⁻¹, respectively. Coordinate covalent bonds were formed between Cu²⁺ and both amino groups and carboxyl groups. In the XPS spectra of the E-CS aerogel and Cu-loaded E-CS aerogel, the interaction between the E-CS aerogel and Cu²⁺ was observable in more detail, as shown in Fig. 7. The total survey scans of the XPS spectra are shown in Fig. 7A. In the XPS spectra of the Cu-loaded E-CS aerogel, new peaks appeared in addition to three intense peaks (O 1s, N 1s, C 1s), which were common for E-CS aerogels; Cu²⁺ was thus successfully adsorbed onto the adsorbent. In Fig. 7B, the peaks at 941.2 eV, 943.8 eV and 962.5 eV indicated the existence of Cu, and the peak at 934.8 eV can be attributed to ionic copper [35]. It was thus inferred that

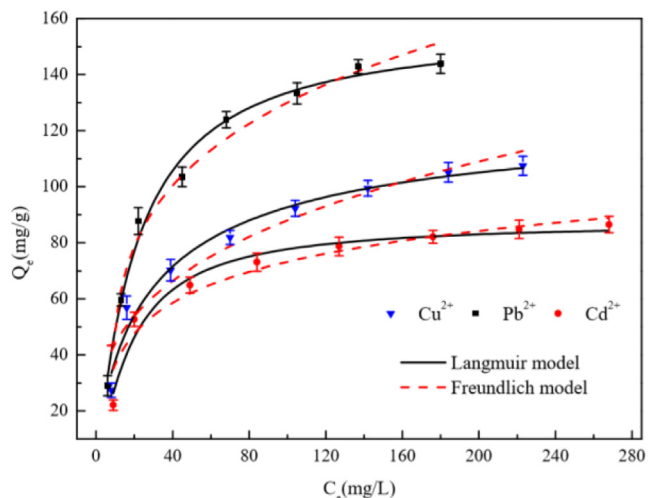


Fig. 6. Adsorption isotherms of E-CS aerogel for Cu²⁺, Pb²⁺ and Cd²⁺. Three parallel tests were performed for each sample.

Table 2

Fitting parameters with different isotherm models for the adsorption process of E-CS aerogel.

Metal ions	Langmuir isotherm model			Freundlich isotherm model		
	$Q_m(\text{mg/g})$	$K_L(\text{L/mg})$	R^2	$K_F(\text{mg}^{1-n}\text{L}^n/\text{g})$	n	R^2
Cu^{2+}	108.14	0.076	0.972	24.73	3.342	0.925
Pb^{2+}	143.73	0.083	0.942	26.65	2.961	0.936
Cd^{2+}	84.62	0.095	0.954	22.17	3.238	0.954

Table 3

Adsorption capacities of various chitosan-based adsorbents for heavy metal ions.

Adsorbent	Process parameters	$Q_{\text{max}}(\text{mg/g})$			Reference
		Cu^{2+}	Pb^{2+}	Cd^{2+}	
Chitosan aerogel	pH = 5.0, T = 298 K	35.08	–	–	[9]
Pollen-chitosan microcapsule	pH = 5.35 for Cd, pH = 5.18 for Cu, T = 298 K	67.10	–	65.98	[10]
Polystyrene-Supported chitosan thin film	pH = 6.0, T = 303 K	99.8	–	–	[11]
Chitosan/polyvinyl alcohol thin membrane	pH = 5.5, T = 313 K	35.1	–	–	[12]
Chitosan-based nanoparticle	pH = 5.5, T = 293 K	–	–	63.5	[13]
Chitosan bead	pH = 6.4, T = 293 K	67.2	55.8	82.6	[14]
Chitosan-modified MnFe_2O_4 nanoparticles	pH = 6.5, T = 298 K	65.1	–	–	[15]
Waste paper/chitosan aerogel	pH = 5.5, T = 298 K	156.3	–	–	[16]
E-CS aerogel	pH = 5.0, T = 298 K	108.14	143.73	84.62	This work

certain copper ions were adsorbed on the adsorbent's surface by physical interaction. Concurrently, the peak at 932.7 eV indicated that some copper ions formed coordination compounds, which could also be observed in other XPS spectra. Before and after Cu^{2+} loading, marked differences were found in the N 1s XPS spectra, as shown in Fig. 7C. The corresponding peaks of C–N and O=C–N of the E-CS aerogel were at 399.4 and 401.5 eV, respec-

tively, which shifted to 399.8 and 402.5 eV after loading Cu^{2+} , respectively. The increased binding energies indicated that the lone pair electrons of N in the amino group interacted with Cu^{2+} , and the electron cloud density around $-\text{NH}_2$ decreased. Fig. 7D shows the O 1s spectra of the E-CS aerogel before and after loading Cu^{2+} . Before loading Cu^{2+} , the binding energies of C–O–H, O=C and C–O–C were 533.1, 532.0 and 531.3 eV, respectively. After

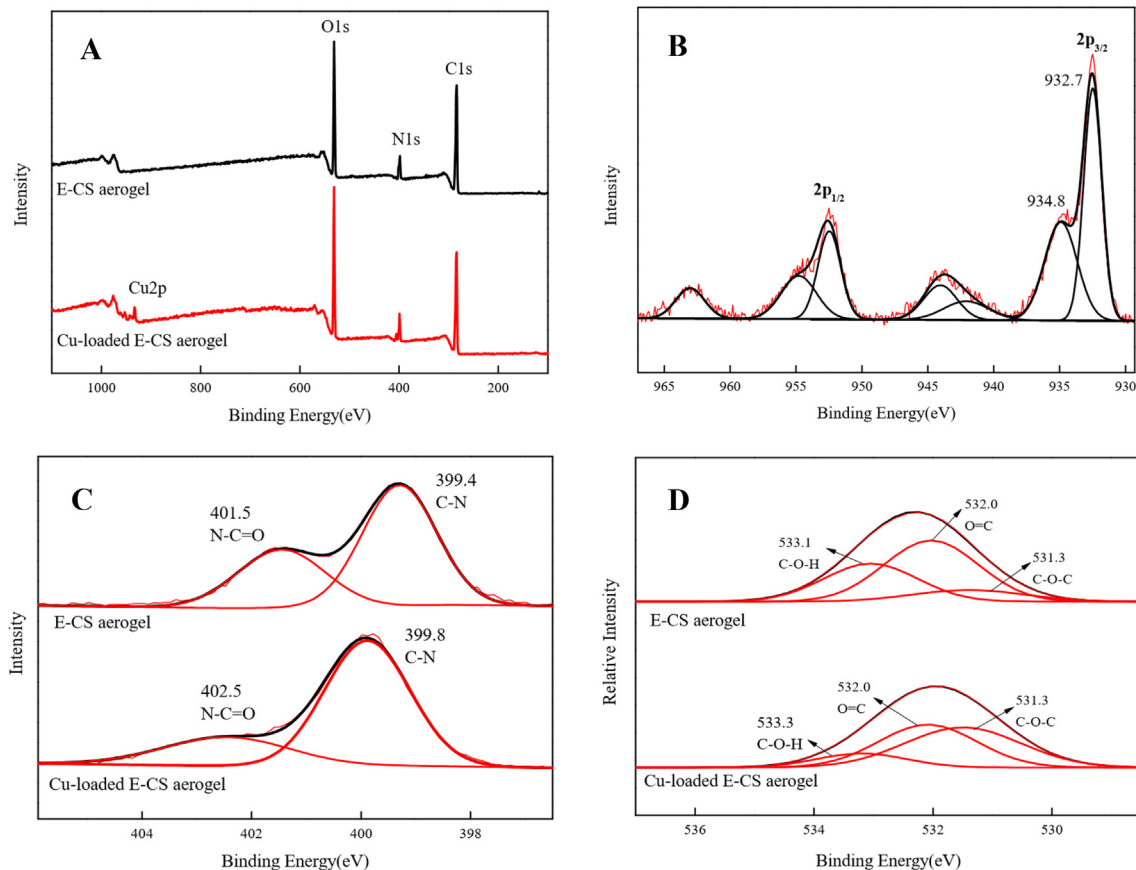


Fig. 7. XPS spectra of the E-CS aerogel and Cu-loaded E-CS aerogel. Total survey scans of XPS spectra (A); Cu 2p spectrum of Cu-loaded E-CS aerogel (B); N 1s spectra of E-CS aerogel and Cu-loaded E-CS aerogel (C); O 1s spectra of E-CS aerogel and Cu-loaded E-CS aerogel (D).

loading Cu^{2+} , the binding energy of C—O—H increased to 533.3 eV. Thus, the lone pair electrons of O in the carbonyl group interacted with Cu^{2+} , and there was a decrease in the density of the electron cloud around C—O—H. Based on these discussion, amino groups and carbonyl groups played coordination roles with Cu^{2+} , and coordination compounds were formed in the Cu-loaded E-CS aerogel.

The adsorption mechanism of the E-CS aerogel includes the comprehensive interaction of Cu^{2+} with the E-CS aerogel, including physical adsorption and combined complexation. The adsorption performance of the E-CS aerogel also benefited from the physical adsorption of the porous structure and from the combined complexation of amino groups and carbonyl groups. To achieve efficient capturing of heavy metal ions with the E-CS aerogel, the combined complexation of amino groups and carbonyl groups had to be present. Thus, even if the porosity decreased from 98.24 to 97.38%, the adsorption performance of the E-CS aerogel was still better than that of the CS aerogel.

4. Conclusions

In this study, a blocky pure chitosan aerogel adsorbent was prepared and enhanced successfully via physical crosslinking and acylation with ethylenediamine tetraacetic anhydride under a relatively easy and efficient ‘top modification’ strategy. This strategy ensured a porous structure of the adsorbent and effectively increased the number of active sites. The capturing ability of the chitosan adsorbent for heavy metal ions was improved dramatically. The saturated adsorption capacities of the E-CS aerogel for Cu^{2+} , Pb^{2+} and Cd^{2+} reached 108.14, 143.73 and 84.62 mg/g, respectively. Investigation of the adsorption mechanism showed that the adsorption process was described accurately by the pseudo second-order kinetic model and Langmuir isotherm model. A comprehensive interaction between Cu^{2+} and the E-CS aerogel occurred, including the combined complexation of amino groups and carbonyl groups and the physical adsorption of porous structures. With their environmental friendliness, good adsorption performances, and easy separation and reusability, E-CS aerogels are attractive for removing heavy metal pollutants from aquatic systems. This ‘top modification’ strategy provides a valuable pathway towards achieving efficient adsorbents for heavy metal ions.

CRediT authorship contribution statement

Shize Li: Conceptualization, Methodology, Investigation, Data curation, Writing - original draft. **Yaping Li:** Validation, Formal analysis. **Zi Fu:** Resources. **Lingbin Lu:** Supervision, Funding acquisition, Writing - review & editing, Project administration. **Jingru Cheng:** Investigation. **Yongsheng Fei:** Investigation.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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